

Robust Watermarking Scheme Using Phase Shift Keying Embedding

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Abstract

Watermarking is a potential method for copyright protection and authentication of multimedia data on the internet. In this paper, a novel watermarking scheme using phase shift keying (PSK) modulation with amplitude boost (AB) and low amplitude block selection (LABS) is proposed to achieve superior performance in terms of robustness and imperceptibility. AB is hired to increase the robustness while LABS is employed to improve the imperceptibility.

In order to demonstrate the effectiveness of the proposed scheme, simulations under various conditions were conducted. The empirical results show that our proposed scheme can sustain most common attacks including JPEG compression, rotating, resizing, cropping, painting, noising and blurring etc.

Keyword: Spread Spectrum (SS), Phase Shift Keying (PSK), Discrete Fourier Transform (DFT), Pseudo-random Number sequence (PN), Joint Photographic Experts Group (JPEG).

1. Introduction

Watermarking techniques can be classified into two categories, one is processed in the spatial domain [1-2] and the other is accomplished in the transform domain. In the transform domain many approaches [3-7] are based on the Discrete Cosine Transform (DCT). Hsu and Wu [4] proposed a scheme by block-based image-dependent permutation of the watermarks in the middle band of the DCT coefficients and obtained good performance. Wu and Hsieh [6] used zerotree structure to embed watermark by rearranging the DCT coefficients in a way similar to the multi-resolution analysis (MRA) of wavelet transforms.

Another method in the transform domain is to hide watermarks in the discrete Fourier transform (DFT) coefficients of the host image. Ruanaidh et al. [13] presented a phase-based method in the DFT

domain and used an optimal detector for watermark recovery. Based on the Fourier-Mellin Transform Ruanaidh and Pun [14] presented a watermarking scheme that achieves rotation, scale and translation (RST) invariant. The scheme achieves robustness while sustains the RST attacks. Premaratne and Ko [15] proposed a new concept for embedding and detecting the watermark in the Discrete Fourier Transform. Since the embedding is independent of the image content, speedy embedding highly suitable for video streams can be achieved. Solachidis and Pitas [16] proposed a watermarking scheme which embeds a circularly symmetric watermark on a ring in the 2D DFT domain. The circularly symmetric watermark was used to solve the rotation invariance problem in the watermark detection in which a correlation operation was used.

In this paper, we proposed a DFT domain watermarking scheme using the phase shift keying (PSK). In our proposed scheme, the watermark bits are first expanded by spread spectrum and then concealed by PSK modulation in the DFT coefficients of the host image. The PSK embedding is employed due to its superior noise immunity. In the PSK embedding, the watermark information is embedded in the phase part of the host image. Thus, the threshold effect in which the quality of the recovered watermark plunges when the amplitudes of the DFT coefficients used for embedding the secret bits are below a threshold value may occur [18]. In this paper, a novel idea combining amplitude boost (AB) and low-amplitude block selection (LABS) is proposed to curb the threshold effect raised by PSK. We demonstrated that by properly combining AB and LABS robustness can be enhanced without sacrificing imperceptibility. Meanwhile, in our scheme neither the original host image nor the original watermark is required during the watermark detection process.

The remainder of this paper is organized as follows. In section 2, the proposed concealing algorithm is presented. The watermark extracting process is presented in section 3. Empirical results are presented in section 4. Finally, section 5 concludes this paper.

2. Concealing Algorithm

A robust watermarking scheme must survive all kinds of attacks, and at the same time sustain the virtual quality of the host image when the watermark is concealed. Besides, security is also an important factor required by a watermarking scheme. In order to construct a superior watermarking scheme, several skills are used in this paper to achieve the goal. The overall concealing process of our proposed scheme is shown in Fig. 1. The watermark W is first transformed to H by toral automorphism (TA) [18] using a pseudo random sequence (PN) generated by a private key to enhance the security. It is then spreaded by spread spectrum (SS) [19-20] to a binary random sequence $\{m\}$. On the other hand, the host image X is transformed to Z by DFT. Then a low amplitude block B is selected from Z using the LABS strategy. Two DFT coefficients ae^{jf} and

ae^{-jf} which from a complex conjugate pair are selected from B for the following AB process in which a is boosted to a' to combat attacks. In the PSK modulation, f is modulated by m into f' . After AB and PSK, $a'e^{jf'}$ and $a'e^{-jf'}$ in which f' contains the secret information are then embedded into Z at the selected block and coefficient pair locations by watermark bits embedding (WBE). The above embedding process is repeated for each secret bit in $\{m\}$ and its corresponding block. The resultant image Z' after embedding is then inversely transformed to obtain the watermarked image R . The details of each block of our embedding algorithm are presented in the following.

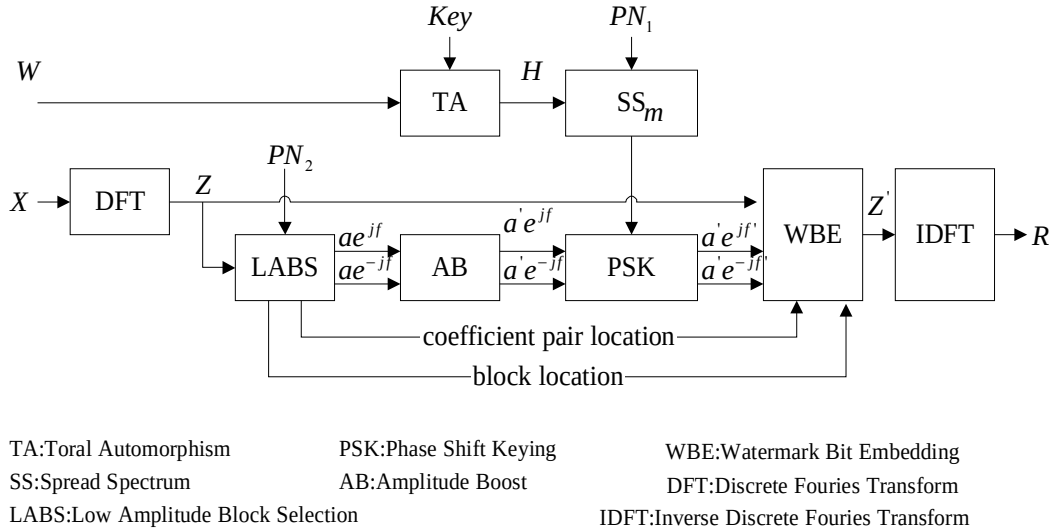


Fig. 1 The flow chart of the proposed embedding process

2.1 Toral Automorphism

For security, the watermark image is pre-permuted into noises by the toral automorphism with a user's key. The toral automorphism scatters the image shape in some iterated operations less than a specified number of times, and will return to the original shape while it is further iterated totally the specified number of times. The specified number is determined by the toral automorphism parameters and the image size. In this paper, the toral automorphism is used to transfer the shape of the original image into chaotic to protect the watermark from being stolen. The transfer function between H

and W is given by

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ k & k+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \pmod{n} \quad (1)$$

Where (x, y) and (x', y') express the pixel locations of W and H respectively, k denotes the control parameter and n denotes the image size, respectively. An example of images transformed by toral automorphism is shown in Fig. 2.



Fig. 2 The images transformed by toral automorphism.

2.2 Spread Spectrum

SS is used to defense noises in many communication systems. A robust watermarking must be able to survive various attacks. In this paper, we hired the SS skill to enhance the robustness of the watermarking scheme. Spread Spectrum expands an information bit into several bits with random values. The expanded bits created when the information bit is high are the inverse of those created when the information bit is low. To randomize the expanded bits, PN sequences are used to accomplish the job. By raster scanning, H is converted into a bit sequence $d(j)$, $j=1,2,3\dots n$. For each $d(j)$, we generate a PN sequence $r_j(i)$, $i=1,2,3\dots l$, l is the length of expansion. By multiplying $d(j)$ by $r_j(i)$, a watermark-bearing bit is chopped up into chips. The expanded bit sequence is given by

$$m_j(i) = d(j) \cdot r_j(i), \quad i=1,2,3\dots l, \quad j=1,2,3\dots n. \quad (2)$$

2.3 Low-Amplitude Block Selection and Amplitude Boost

In this paper, two complementary strategies, amplitude boost (AB) and low-amplitude block selection (LABS) are employed to design a novel embedding scheme using the PSK modulation. The amplitude boost is a skill used to enhance robustness, while the low-amplitude block selection is used to preserve the imperceptibility as much as possible. The low amplitude block selection strategy selects blocks of low amplitude to embed the secret bits.

To make a watermarking scheme robust, a good strategy is to embed the watermark bits into the significant portion of the host signal, because this portion of the host data is highly sensitive to alteration. A watermark concealed in the high or middle frequency bands is easier to be removed or altered without affecting the host image by attacks [22]. In this paper, we embedded the secret data in lower frequency bands to enhance the robustness. The DFT of a block image B of size 8×8 is given by

$$B(u, v) = \frac{1}{\sqrt{64}} \sum_{x=0}^7 \sum_{y=0}^7 b(x, y) e^{-j2\pi xu/8 - j2\pi yv/8}, \quad 0 \leq u \leq 7, 0 \leq v \leq 7 \quad (3)$$

with the inverse transform given by

$$b(x, y) = \frac{1}{\sqrt{64}} \sum_{u=0}^7 \sum_{v=0}^7 B(u, v) e^{-j2\pi xu/8 - j2\pi yv/8}, \quad 0 \leq x \leq 7, 0 \leq y \leq 7 \quad (4)$$

In the PSK modulation, the watermark information is contained in the phase of the DFT coefficients. When a watermarked image is attacked, the DFT coefficients of the watermarked image are altered, which produces distortion when a watermark is recovered from the attacked watermarked image.

DFT coefficient due to an additive Gaussian noise is inversely proportional to the amplitude of the coefficient. In order to avoid the large distortion caused by DFT coefficients of small amplitudes, a novel strategy called AB is employed.

The AB strategy is to boost the amplitude of a selected DFT coefficient to a threshold value th when its value is below th so that the phase distortion under attacks can be keep below a certain level. In other words, after AB the amplitude of all the DFT coefficients used for embedding the secret bits are all above th . That is,

$$a' = \begin{cases} th & , \quad \text{if } a \leq th \\ a & , \quad \text{if } a > th \end{cases} \quad (5)$$

2.5 PSK Embedding

In the PSK modulation, the phase f is modified into f' according to

$$f' = a(f, s) + sb(m) \quad (6)$$

where s denotes the watermark strength factor, a and b denote the offset function and the embedding function respectively given by

$$a(f, s) = 2psi, \quad \text{if } |f - 2psi| \leq |f - 2psj|$$

$$\text{for } j = 0, 1, 2, \dots, \frac{1}{s} - 1, j \neq i$$
(7)

$$b(m) = \begin{cases} p/2, & \text{if } m = 0 \\ -p/2, & \text{if } m = 1 \end{cases}$$
(8)

Fig. 3 shows the coefficient pair selected within a block for embedding the secret bit and Fig. 4 shows the signal constellation of the PSK modulation for $s=1$, $s=1/2$ and $s=1/4$.

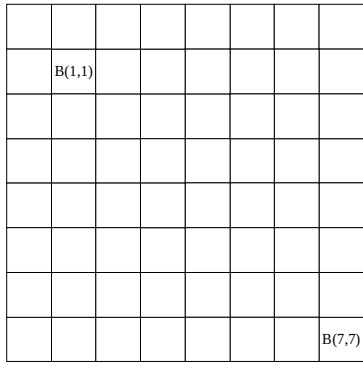


Fig. 3 The coefficient pair selected within a block for embedding the secret bit.

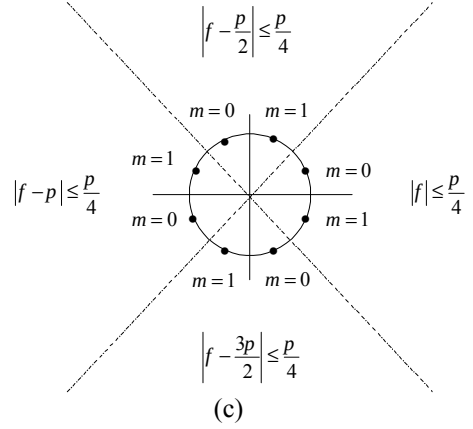
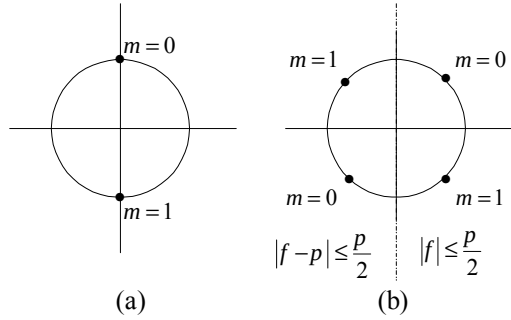


Fig. 4. The signal constellation of the PSK for (a) $s=1$, (b) $s=1/2$, and (c) $s=1/4$

3. Watermark Extraction Algorithm

A process which inverses the concealing process is used to recover the original watermark. The flow chart of the recovering process is shown in Fig. 5. The watermarked image R is transformed to Z' by DFT. The same PN sequence PN_2 used in the concealing process is used to select the embedded blocks from Z' for the PSK demodulation. In the PSK demodulation, the secret bit m' is extracted from the phase f'' of the selected DFT coefficient for each selected block. After all the secret bits are extracted from the PSK demodulation, they are contracted by inverse spread spectrum (ISS) and rearranged into the two-dimensional image H' . By passing H' through the inverse toral automorphism (ITA), the recovered watermark W' is obtained.

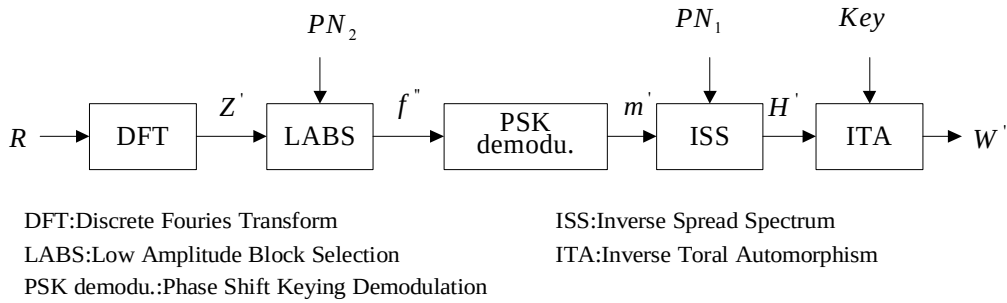


Fig. 5 The flow chart of watermark detection.

3.1 PSK Demodulation

In the process of the PSK demodulation the angle f'' is processed to recover the embedded secret bit

$$m' = \begin{cases} 0, & \text{if } |f'' - f_{1i}| \leq |f'' - f_{jk}|, \quad j=1,2; k=0,1,2,\dots,\frac{1}{s}-1, \quad k \neq i, \quad \text{when } j=1 \\ 1, & \text{if } |f'' - f_{2i}| \leq |f'' - f_{jk}|, \quad j=1,2; k=0,1,2,\dots,\frac{1}{s}-1, \quad k \neq i, \quad \text{when } j=2 \end{cases} \quad (9)$$

3.2 Inverse Spread Spectrum

Since SS expands watermark bits before embedding, they will be reconstructed by contraction. A bit $d'(j)$ in H' is obtained by contracting its expanded sequence $m'_j(i)$, $i=1,2,3\dots l$, using $r_j(i)$ generated by PN_1 . The value of $d'(j)$ is determined by

$$d'(j) = \begin{cases} 1, & \text{if } \sum_i m'_j(i) \oplus r_j(i) < \frac{l}{2} \\ 0, & \text{if } \sum_i m'_j(i) \oplus r_j(i) \geq \frac{l}{2} \end{cases} \quad j=1,2,\dots,n \quad (10)$$

Where $\oplus$ denotes the XOR operator.

4. Experimental Results

Imperceptibility is an important factor for watermarking. In this paper we employ the PSNR to indicate the degree of transparency. The PSNR of R is given by

$$PSNR = 10 \log_{10} \frac{255^2}{\frac{1}{N \times N} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (X(i,j) - R(i,j))^2} \quad (11)$$

Where $X(i,j)$ and $R(i,j)$, are the gray values at (i,j) of the host image X and the watermarked image R of size $n = N \times N$ respectively. The watermark similarity measurement is dependent on factors such as the knowledge of the experts, the experimental conditions, etc. Therefore a quantitative measurement is necessary to provide fair

judgment of the extracted fidelity. In this paper, we use the normalized correlation (NC) between the reference watermark W and the extracted watermark W' as the similarity measurement.

$$NC = \frac{\sum_i \sum_j [W(i,j)W'(i,j)]}{\sum_i \sum_j [W(i,j)]^2}, \quad (12)$$

NC is normalized by the reference watermark energy to give unity as the peak correction.

The images Lena (512×512) and Baboon (512×512) are used in simulation for demonstrating the performance of the proposed scheme. The logo image, National Chinyi Institute of Technology (32×32) and publications of department of electronic engineering (32×32) in Chinese were used as the watermark. The block size used is 8×8 . The values of parameters used in the simulations are: $k=5$, $s=\frac{1}{2}$, $l=3$, $b_1=3$, $b_2=1$, $th=9$ for the Baboon image and $th=13$ for the Lena image. The number of the secret bits after SS expansion is equal to $32 \times 32 \times l = 3072$. The total number of blocks is equal to $(512 \times 512) \div (8 \times 8) = 4096$. The number of blocks selected for concealing the secret bits is equal to $4096 \times \frac{b_1}{b_1 + b_2} = 3072$ which is equal to the

total number of the secret bits. The embedding scheme with AB and LABS is called WABLABS was used to test.

Fig. 6 shows the original image and the watermarked images of the WABLABS embedding scheme. The PSNR of the watermarked images are 35.35 and 34.12 for the Lena and Baboon images respectively. From Fig. 6, one could hardly perceive

the difference between the watermarked image and the original image. Figs. 7 and 8, show the comparison results from the JPEG lossy compression attacks using Photoshop 6.0 with quality levels from 0 to 12 for images Lena and Baboon respectively. Tables 1 and 2 show the empirical comparison results under the rotating, resizing, cropping, painting, noising and blurring attacks for the Lena and Baboon images respectively.

The normalized correlation and the extracted watermarks corresponding to Fig6 are shown in tables 3 and Fig. 7, as well as table 4 and Fig. 8 respectively. From the test results, the values of

NC are all above 0.88 and the extracted watermarks are clearly identified by the human vision.

Fig. 9 (a)-(f) show the 90-degree rotated, the resized, the cropped, the painted, the noised, and the blurred watermarked images and Fig. 9(g)-(l) show their corresponding extracted watermarks using the WABLABS scheme for the Lena image. The corresponding numbers of error bits of the extracted watermarks are listed in table 1. The visual quality of the extracted watermarks revealed in Fig. 9 (g)-(l) demonstrated that our proposed scheme can sustain all the above attacks.



Fig. 6 (a) the original image of Lena (b) the embedded image of Lena with the PSNR=35.35 (c) the original image of Baboon (d) the embedded image of Baboon with PSNR=34.12



Fig. 7 The extracted watermarks corresponding to JPEG compression attacks for the image Lena by photoshop 6.0.



Fig. 8 The extracted watermarks corresponding to JPEG compression attacks for the image Baboon by photoshop 6.0.



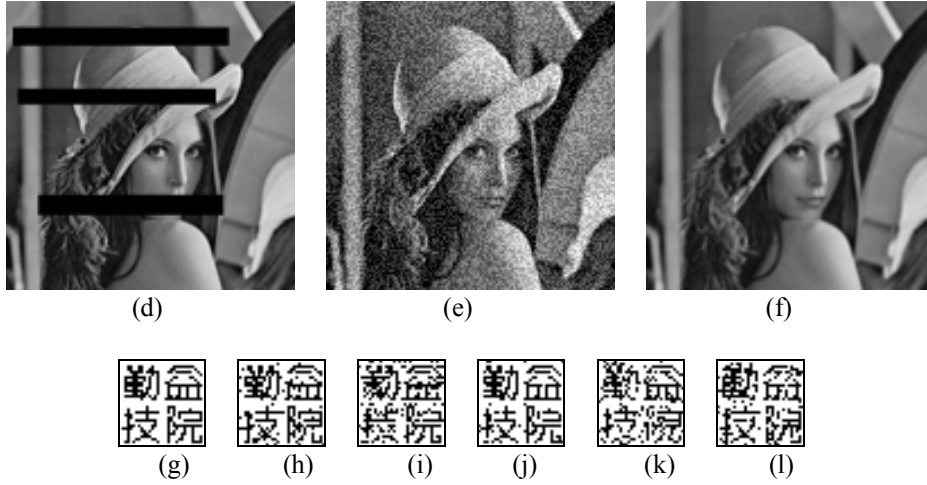


Fig. 9. The attacked watermarked images (a) – (f), (a) the 90-degree rotated (b) the resized (c) the cropped (d) the painted (e) the noised, and (f) the blurred. The corresponding extracted watermarks (g) – (l), (g) the 90-degree rotated (h) the resized (i) the cropped (j) the painted (k) the noised, and (l) the blurred.

Table 1 Comparison of the number of the error bits of the recovered watermark under various attacks for the Lena image.

Item	Parameter	WABLABS PSNR=35.35
Rotating	90 degree rotated	0
Resizing	From 512×512 to 192×192	52
Cropping	Cropping a quarter image	107
Painting	Painting three bars	29
Noising	15.06% noises contamination	103
Blurring	Gaussian blurring ratio 2.0	94

Table 2 Comparison of the number of the error bits of the recovered watermark under various attacks for the Baboon image.

Item	Parameter	WABLABS PSNR=34.12
Rotating	90 degree rotated	0
Resizing	From 512×512 to 192×192	108
Cropping	Cropping a quarter image	93
Painting	Painting three bars	33
Noising	15.06% noises contamination	97
Blurring	Gaussian blurring ratio 1.4	88

Table 3. The NC values corresponding to the number of error bits for the image Lena under JPEG compression attacks by photoshop 6.0.

Quality level	0	1	2	3	4	5	6	8	10	12
Error bits no	70	2	0	0	0	0	0	0	0	0
NC	0.92	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Table 4. The NC values corresponding to the number of error bits for the image Baboon under JPEG compression attacks by photoshop 6.0.

Quality level	0	1	2	3	4	5	6	8	10	12
Error bits no	89	17	1	0	0	0	0	0	0	0
NC	0.88	0.95	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0

5. Conclusions

Digital watermarking is a potential method to discourage unauthorized copying or attest origin of

digital data that includes audio, video and images. In this paper we present a robust watermarking scheme for still images using PSK with amplitude boost and low amplitude block selection. The amplitude boost strategy is used to enhance the robustness and the low amplitude block selection strategy is used to reduce the degradation of the host image caused by watermark concealing. Empirical results show that the proposed scheme can sustain attacks like JPEG lossy compression, rotation, resizing, cropping, painting, noising and blurring.

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